

Evaluation of LLMs on Math Problems

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Motivation

- ★ Growing reliance on LLMs for mathematical reasoning in education and professional work
- ★ Existing benchmarks narrow in scope and focused mainly on final answers
- ★ Risk of misleading or pedagogically harmful explanations
- ★ Need to understand when LLMs support learning vs. when human oversight is essential

Dataset

StackMathQA: mathematical questions and answers sourced from the Stack Exchange network
Ranges from enthusiast discussions to expert-level math problems
100 questions sampled, labeled into 5 types: *numeric*, *proof*, *yes/no*, *concept*, *open*
Each question has multiple human-written answers used as reference

Models

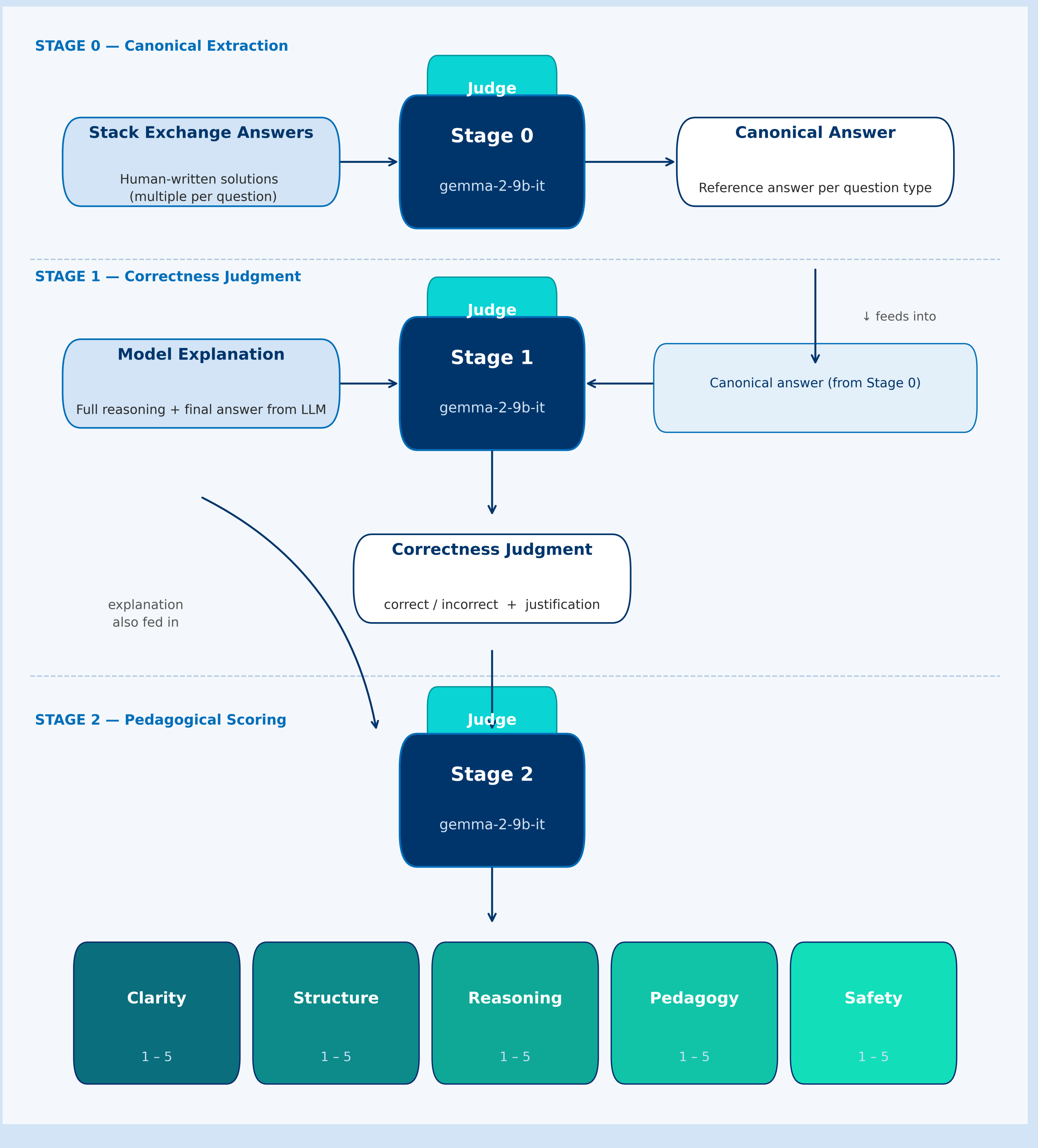
Solver models (same prompt for all):

- ★ Meta-Llama-3.1-8B-Instruct: general baseline
- ★ Qwen2.5-7B-Instruct: strong general 7B
- ★ Qwen2.5-Math-7B-Instruct: math-specialized 7B

Judge model:

- ★ google/gemma-2-9b-it: strong in summaries and reasoning

Evaluation Pipeline



Method

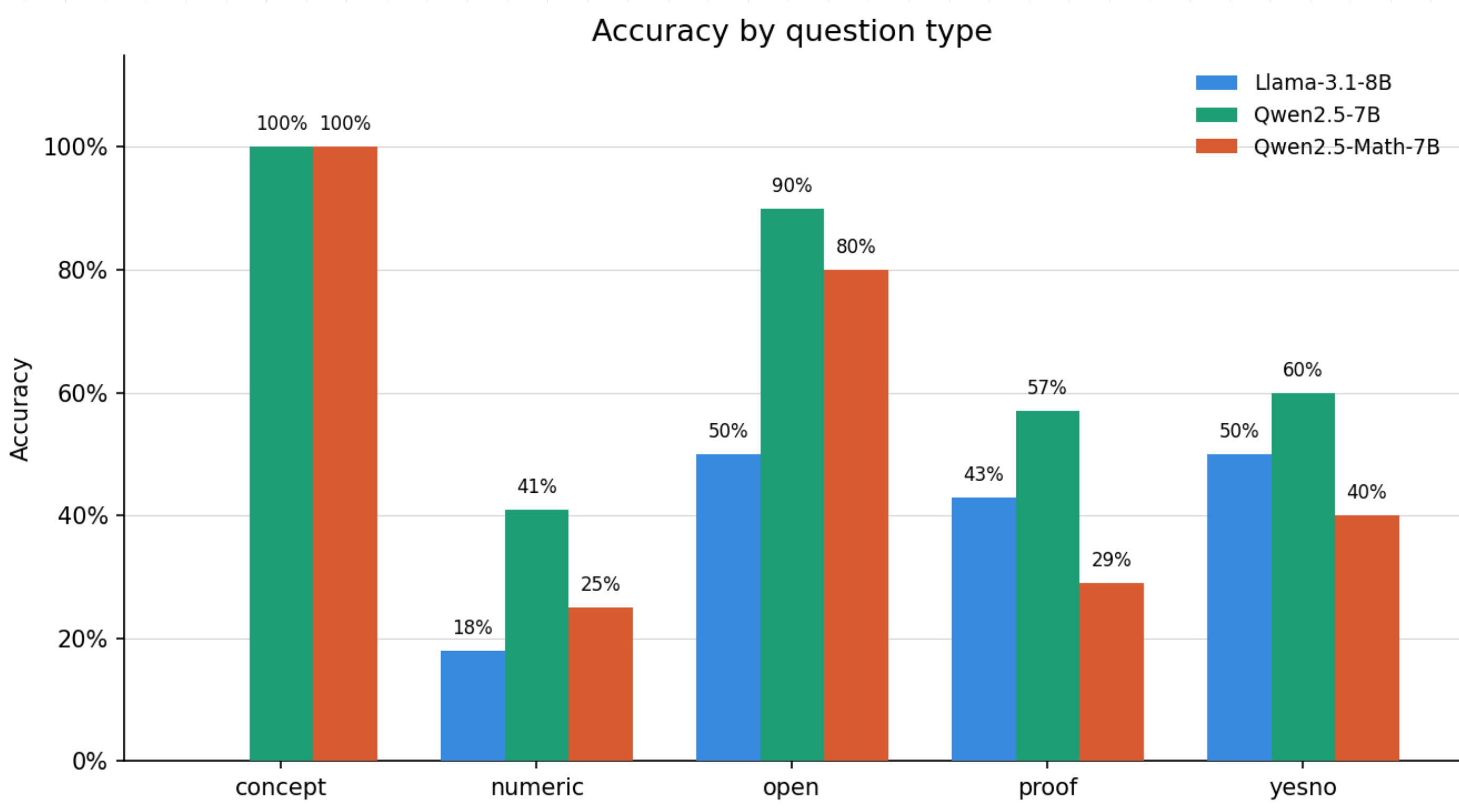
Evaluation prompt (identical for all models):

- ★ Models instructed to **show reasoning step by step** (max 8–10 sentences)
- ★ Must end with Final answer: `<answer>`
- ★ If insufficient information: instructed to say so rather than guess

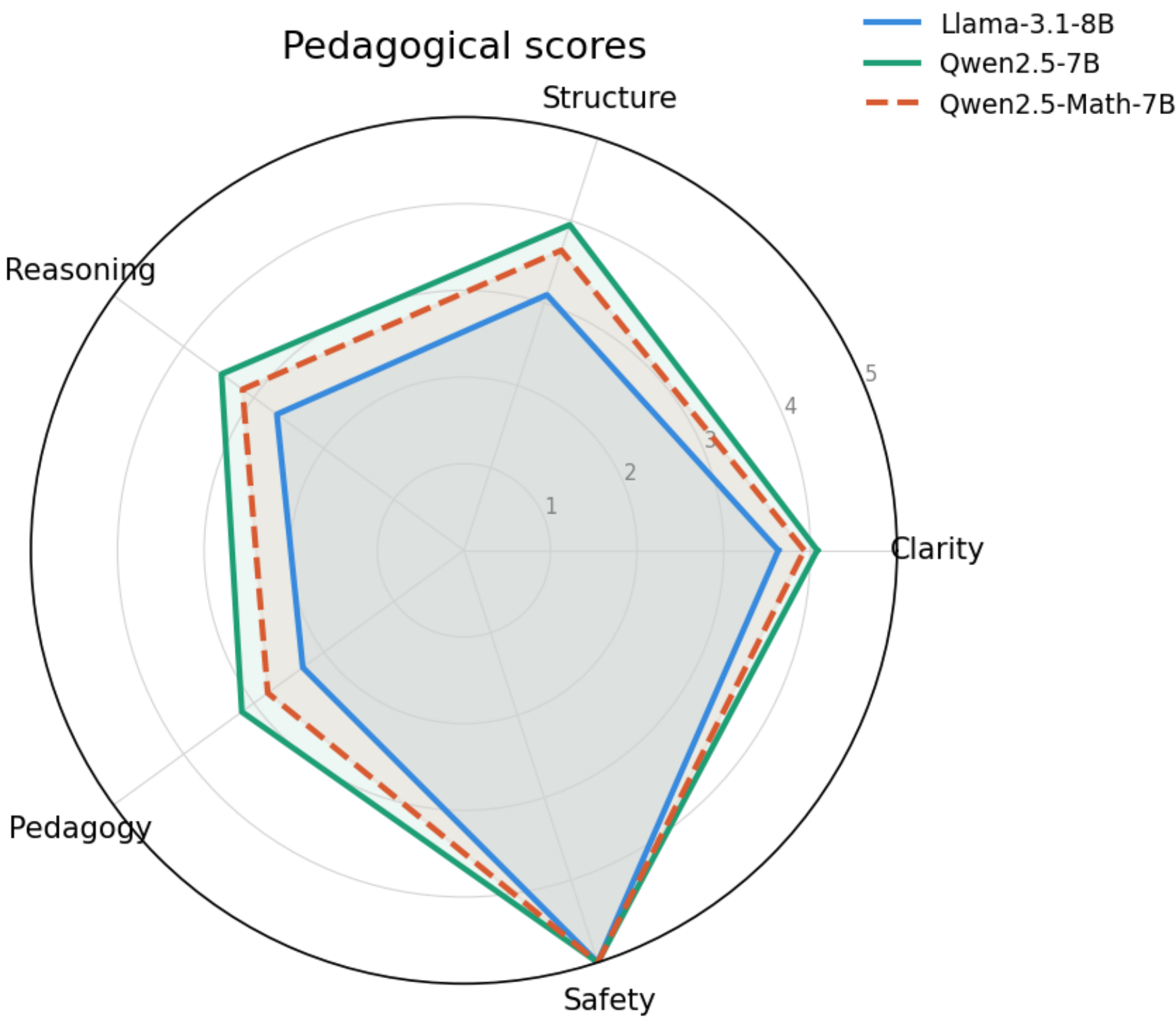
Metrics:

- ★ **Accuracy:** correctness of final answer vs. canonical reference
- ★ **Pedagogical score:** clarity, structure, reasoning, pedagogy, safety (1–5 scale)

Results: Accuracy by Question Type



Results: Explanation Quality Score



Results: Overall Summary

